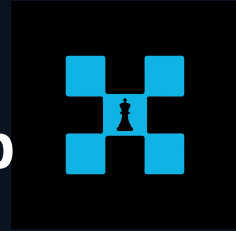


# ChessEver Desktop Roadmap



Turning the existing Flutter mobile app into a professional Windows-first desktop workspace that can credibly rival ChessBase.

**Prepared from pubspec.yaml, pubspec.lock, local architecture, and official ChessBase sources**

**Date: May 1, 2026**

Audience: product, design, Flutter, backend, data, and chess-domain engineering leads.

Scope: this is a build roadmap and technical due-diligence document, not a commitment to clone proprietary ChessBase formats or protected behavior.

# 1. Executive Decision

## Recommendation

Keep Flutter and reuse the pure chess/domain stack, but treat the desktop app as a product rewrite at the shell, interaction, storage, and engine layers. The current app can become the base of a professional desktop product only after platform adapters, a desktop database/indexing layer, a process-based engine service, and a command-driven workspace UI are introduced.

### Reusable now

Dart chess model, PGN parsing/export, notation tree with variations, Gamebase API models, Riverpod providers, Supabase library/sync concepts, Inter assets, and existing board/game tests.

### Must be redesigned

Navigation, sheets, phone/tablet scaling, route model, app startup, keyboard/mouse interactions, desktop menu bar, workspace layout, and professional data workflows.

### Must be replaced

Mobile-only Stockfish plugin path, mobile SQLite path, push/attribution/app-review/update assumptions, mobile file-share intents, and Windows monetization/auth gaps.

### Strategic advantage

Existing Flutter mobile code, Gamebase service, Supabase accounts, and PGN/analysis work can become a cross-device ecosystem faster than a pure native rewrite.

## What 'rival ChessBase' means in practice

ChessBase is not just a chessboard UI. It is a professional workspace for databases, preparation, reference search, annotated games, engine analysis, player reports, opening reports, cloud/account workflows, and high-volume data updates. The desktop version of ChessEver must therefore behave like a productivity tool: dense, keyboard-first, multi-pane, fast on large datasets, and reliable under long analysis sessions.

## Current feasibility summary

| Area          | Current state                                                   | Desktop readiness              | Action                                                                            |
|---------------|-----------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------|
| Core chess    | dartchess, chess, chessground, PGN import/export, notation tree | Medium-high                    | Extract into a pure, testable domain package and add conformance tests.           |
| Desktop shell | Mobile Material routes, bottom sheets, hamburger/nav flows      | Low                            | Build a Windows-first workspace shell with panes, menus, commands, and shortcuts. |
| Engine        | stockfish package and singleton tuned for mobile FFI lifecycle  | Low                            | Replace desktop engine with UCI process manager and bundled engines.              |
| Local storage | sqlite AppDatabase used for key-value/cache/list tables         | Low for Windows                | Move to sqlite3/drift/sqlite_common_ffi and add game/position indexes.            |
| Cloud/backend | Supabase, Gamebase service, Lichess/FIDE requests               | Medium                         | Keep, but add desktop auth, offline cache, sync queue, and API rate/error policy. |
| Testing       | Unit/widget tests plus Patrol mobile E2E                        | Medium for mobile, low desktop | Add desktop widget/golden/integration tests and Windows CI.                       |

## 2. Evidence From Current Repo

Read from the current worktree: pubspec.yaml, pubspec.lock, .dart\_tool/package\_config.json, local pub-cache plugin metadata, main.dart, repository/gamebase, repository/library, repository/sqlite, chessboard analysis/notation, PGN intake, responsive/tablet helpers, tests, and platform folders. The project is chesserver2 version 14.0.0+1400, with Dart SDK constraint ^3.7.2.

| Finding                          | Evidence                                                                                                                                   | Implication                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Flutter desktop folders exist    | windows/, macos/, linux/, and web/ are present.                                                                                            | The repo has platform scaffolding, but that is not the same as a production desktop app. |
| Toolchain is modern              | Observed Flutter 3.41.9 stable and Dart 3.11.5 on this machine.                                                                            | Modern Flutter desktop APIs are available, but dependency/platform gaps still matter.    |
| State model                      | hooks_riverpod 2.6.1, flutter_hooks, Riverpod providers across screens.                                                                    | Good basis for splitting desktop workspace state from mobile navigation state.           |
| Chess model exists               | ChessGame.fromPgn, NotationTreeBuilder, exportGameToPgn, move variations, comments, NAGs, clocks, eval comments.                           | Keep and harden as a shared domain layer.                                                |
| Gamebase exists                  | Dio repository for move aggregates, players, games, eval, global search; local doc says backend is Hono.js/TypeScript + PostgreSQL/Prisma. | This is the seed of reference search and player preparation.                             |
| Library exists                   | Supabase folders, saved analyses, shared books, TWIC synthetic folder, import flows.                                                       | Good concept, but desktop needs local-first database semantics.                          |
| Mobile startup assumptions       | main.dart initializes Sentry, Supabase, OneSignal, RevenueCat, Appsflyer, deep links, PGN share intents, orientation controls.             | Desktop needs platform-specific startup gates and adapters.                              |
| Responsive model is phone/tablet | ResponsiveHelper uses 393x852 base size and DeviceType phone/tablet.                                                                       | Professional desktop cannot be a scaled tablet layout.                                   |

### Direct dependency shape

| Group                | Direct package count | Desktop consequence                                                            |
|----------------------|----------------------|--------------------------------------------------------------------------------|
| backend/network      | 6                    | Reusable with desktop auth and cache policy validation.                        |
| chess                | 4                    | Strong domain starting point; Stockfish desktop path is the main gap.          |
| i18n/assets          | 12                   | Mostly reusable; audio package has desktop metadata.                           |
| modeling/build/test  | 15                   | Keep; add desktop-specific test harnesses.                                     |
| monetization/auth    | 4                    | RevenueCat and Google Sign-In do not cover Windows in local metadata.          |
| observability/growth | 8                    | Sentry supports desktop; growth SDKs are mostly mobile-oriented.               |
| other                | 3                    | Validate individually.                                                         |
| platform/mobile      | 7                    | Most should be disabled or replaced in desktop builds.                         |
| state/ui             | 11                   | Useful for shared UI, but desktop layout should be rebuilt rather than scaled. |
| storage/files        | 10                   | File picker is desktop-ready; sqlite and share-intent flows need replacement.  |

Full direct dependency readiness is in Appendix A.

### 3. ChessBase Baseline

Official ChessBase pages available on May 1, 2026 position the product as a Windows desktop database and preparation suite, with current packages centered around ChessBase '26 and Mega Database 2026. The important lesson is not to copy the UI. The lesson is the professional job-to-be-done: prepare against players, search huge game sets, analyze deeply, annotate efficiently, and train from the results.

| ChessBase capability/claim from official sources                                                                                                                                                 | Product implication for ChessEver                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| ChessBase '26 Mega Package is Windows 10 or higher, published Nov 2025, with minimum 4 threads/8 GB RAM and recommended i7/Ryzen 8 threads/16 GB RAM plus SSD for professional database work.    | Target Windows first. Define performance budgets around SSD-backed local indexes and large databases, not mobile cache assumptions.      |
| Mega Database 2026 lists more than 11.7 million games from 1475-2025, more than 114,000 annotated games, weekly update service of more than 5,000 games, and a player lexicon with names/photos. | Chessever needs a data strategy: Gamebase scale, update cadence, local indexing, annotated-game pipeline, and player identity model.     |
| ChessBase 18/26 pages emphasize opening reports, reference search, strategic themes, style reports, blunder reports, online-game access, cloud engines, and mobile/cloud preparation access.     | The desktop app must combine search, prep reports, engine analysis, and cloud sync into one workflow instead of separate mobile screens. |
| ChessBase highlights player preparation, tournament work, training material, publishing, and professional use from youth players to grandmasters.                                                | Chessever should optimize for coaches, titled players, clubs, and ambitious tournament players, not only live-event browsing.            |

#### Differentiation angle

Do not try to beat ChessBase by matching every legacy feature first. Start with a cleaner modern workflow: cloud-linked mobile and desktop accounts, fast Gamebase reference search, beautiful board/notation ergonomics, modern engine analysis, and a professional database shell that feels less legacy and more focused.

## 4. Product Architecture Target

### Desktop architecture modules

| Module                | Owns                                                                                                            | Why it matters                                                                |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| core_chess            | FEN/PGN parsing, game tree, notation, variations, comments, NAGs, move navigation, export.                      | Keeps chess correctness independent from mobile or desktop UI.                |
| desktop_shell         | Windows/macOS/Linux app shell, menu bar, commands, shortcuts, docked panes, document tabs, command palette.     | Makes the app feel native to keyboard/mouse professionals.                    |
| engine                | UCI process manager, engine profiles, MultiPV, analysis queue, cancellation, cloud-engine adapter, engine logs. | Replaces mobile Stockfish lifecycle with desktop-grade long-running analysis. |
| data_index            | Local SQLite/FTS schema, PGN import workers, position index, game/player/event/search indexes, sync status.     | The core requirement for ChessBase-grade database work.                       |
| sync_services         | Supabase auth, library sync, Gamebase API, entitlements, share/book subscriptions, conflict policy.             | Keeps existing cloud value while enabling offline work.                       |
| platform_adapters     | Database, files, auth redirects, notifications, updates, purchases, analytics, engine binaries.                 | Allows mobile and desktop to share domain code without platform crashes.      |
| professional_features | Reference search, opening reports, player dossiers, repertoire tools, blunder/style reports, training mode.     | This is where ChessEver competes, not merely where it runs.                   |

### Key interfaces to add before UI work expands

| Interface     | Desktop implementation                                                                 | Mobile implementation                                           |
|---------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| LocalDatabase | sqlite3/drift or sqflite_common_ffi with WAL, FTS5, workers, and migrations.           | Current sqflite-backed cache or migrated shared implementation. |
| EngineRunner  | Launch bundled Stockfish-compatible executable via dart:io Process; parse UCI streams. | Existing plugin/FFI path until mobile is also migrated.         |
| FileIntake    | Open dialogs, drag/drop, file associations, recent files, watched folders.             | file_picker plus receive_sharing_intent.                        |
| Entitlements  | Stripe/web checkout or RevenueCat-compatible desktop path; cached license state.       | Current RevenueCat mobile path.                                 |
| AuthLauncher  | Desktop OAuth redirect/deep link loopback, browser launch, secure token cache.         | Current Supabase mobile OAuth/deep link handling.               |
| AppLifecycle  | Window close, document autosave, engine shutdown, tray/background policy.              | Mobile background/resume policy.                                |

## 5. Desktop UX Rewrite

The desktop app should start as the working environment, not a landing page or mobile home screen. First viewport: board, notation, engine/reference panes, database explorer, and command surfaces. Every action should be reachable by mouse and keyboard.

| Surface           | Desktop behavior                                                                                                                  | Current code impact                                                            |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Menu bar          | File, Edit, View, Game, Engine, Database, Training, Window, Help. Include accelerators and disabled states.                       | Add shell-level command registry; mobile routes should not own global actions. |
| Workspace         | Dockable or fixed professional panes: board center, notation right, engine/reference bottom or side, database/player search left. | Replace bottom-sheet-heavy flows with panes/dialogs sized for desktop.         |
| Document tabs     | Open multiple games/databases/prep boards. Dirty state, save prompts, recent files.                                               | Current Navigator stack must become document/workspace state.                  |
| Board interaction | Drag/drop, click-to-move, premove optional, right-click arrows/highlights, wheel navigation, hover square metadata.               | chessground likely helps, but desktop pointer/focus testing is required.       |
| Notation          | Selectable move tokens, keyboard traversal, variation folding, inline comments, NAG palette, engine annotation insertion.         | Existing notation tree is a strong base; UI must become editable and dense.    |
| Search            | Global command/search palette plus database-specific filters. Results in tables with sort, resize, open-in-board.                 | Reuse Gamebase/global search providers; build desktop table components.        |
| Dialogs           | Use modal dialogs for rare confirmations and inspectors for persistent editing. Avoid mobile sheets for primary work.             | Smooth sheets should be mobile-only or rare on desktop.                        |

### Keyboard and mouse command baseline

| Command              | Windows/Linux         | macOS          | Notes                                       |
|----------------------|-----------------------|----------------|---------------------------------------------|
| Command palette      | Ctrl+K                | Cmd+K          | Search actions, games, players, settings.   |
| Open PGN/database    | Ctrl+O                | Cmd+O          | File dialog plus drag/drop.                 |
| Save analysis        | Ctrl+S                | Cmd+S          | Save active document, not only cloud sheet. |
| Find/search          | Ctrl+F                | Cmd+F          | Scoped to active pane.                      |
| Move previous/next   | Left/Right            | Left/Right     | Works when board/notation focused.          |
| Variation navigation | Up/Down or Alt+Arrows | Option+Arrows  | Move between sibling lines.                 |
| Start/end            | Home/End              | Home/End       | Go to start/end of active line.             |
| Flip board           | F                     | F              | Visible state in toolbar.                   |
| Toggle engine        | Space or E            | Space or E     | Must not conflict with text fields.         |
| Annotate             | !, ?, Ctrl+1..6       | !, ?, Cmd+1..6 | NAG insertion and quick comments.           |

## 6. Data And Search Requirements

### Highest-risk area

A beautiful desktop UI without a serious local database/index layer will not rival ChessBase. The current SQLite usage is a mobile key-value/cache/list replacement. Professional desktop needs game, position, player, event, annotation, and search indexes, with import jobs and offline-first behavior.

| Table/index    | Fields and indexes                                                                                                | Purpose                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| games          | id, source, headers, pgn_raw, result, date, eco, event_id, white_id, black_id, folder_id, updated_at, sync_status | Canonical game records and user library entries.              |
| game_moves     | game_id, ply, san, uci, fen, comments, nags, eval, variation_path                                                 | Editable notation tree and fast board navigation.             |
| positions      | position_hash, fen4, game_id, ply, next_uci, side_to_move, player_to_move, rating, result, year, time_control     | Reference search and move statistics.                         |
| players        | id, fide_id, normalized_name, title, fed, ratings, aliases, online_accounts                                       | Player prep and de-duplication.                               |
| events         | id, name, site, start_date, category, source                                                                      | Tournament and database filters.                              |
| analysis_cache | fen4, engine_id, depth, multipv, pvs, created_at                                                                  | Avoid recomputing common positions.                           |
| folders/books  | folder tree, ownership, share/subscription metadata, sync version                                                 | Bridge current Supabase library model to local-first desktop. |
| fts indexes    | FTS5 on player names, events, comments, openings, PGN headers                                                     | Professional instant search.                                  |

### Import pipeline

| Step           | Behavior                                                                                                                                                     |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File discovery | Open file, drag/drop, folder import, recent files, watched download folder, and .pgn file association.                                                       |
| Parse          | Use existing PGN parser as seed, then harden for multi-game files, encodings, malformed tags, variants, comments, clocks, evals, NAGs, and very large files. |
| Normalize      | Canonical names, dates, results, ECO/opening fields, FIDE IDs, player aliases, source fingerprints, duplicate detection.                                     |
| Index          | Batch write games and positions, compute fen4/position hashes, populate FTS and compound indexes.                                                            |
| Review         | Show import report: imported, skipped, duplicates, failed games, warnings.                                                                                   |
| Sync           | Queue cloud sync after local save; never block desktop work on network.                                                                                      |

## 7. Engine And Analysis System

The desktop app should use Stockfish-compatible UCI engines as external processes. This matches desktop expectations, allows engine management, makes engine upgrades easier, and avoids relying on a mobile-only plugin.

| Requirement        | Implementation note                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| Engine manager     | Detect bundled engine, let users add engines, configure threads/hash/skill/MultiPV/tablebase path, and store profiles.    |
| UCI process runner | Use dart:io Process, parse stdout/stderr, send uci/isready/position/go/stop/quit, expose structured stream updates.       |
| Job queue          | Prioritize visible board, cancel stale jobs, allow background infinite analysis, and prevent cross-document result leaks. |
| Analysis panes     | Show eval bar, PV table, depth/nodes/NPS, best move, threats, blunders, and copyable lines.                               |
| Cloud engines      | Keep future adapter for paid remote compute; current Gamebase eval endpoint can seed cloud eval fallback.                 |
| Persistence        | Cache engine evals by fen4 + engine profile + depth/MultiPV; invalidate when engine/profile changes.                      |
| Safety             | Kill orphaned processes on close, cap CPU defaults, expose pause button, keep logs for support.                           |

### Professional analysis features to sequence

| Feature          | MVP                                                           | Later                                                            |
|------------------|---------------------------------------------------------------|------------------------------------------------------------------|
| Instant analysis | Evaluate current board with 3 PVs and stable cancellation.    | Infinite analysis, background tabs, batch annotate game.         |
| Blunder report   | Classify centipawn loss and missed tactics from local engine. | Calculation Elo, phase-specific weakness profile, training plan. |
| Opening report   | Gamebase/local position stats for current line.               | Plans, piece paths, rating-band stats, typical tactics.          |
| Player dossier   | Opening choices, score by color/time/year/opponent level.     | Style report, account matching, prep recommendations.            |
| Tablebases       | Optional path configuration.                                  | Syzygy integration and endgame report UI.                        |



## 8. Platform And Dependency Work

The repo can keep one codebase, but desktop must not initialize mobile-only services or rely on mobile-only plugins. The safest pattern is interface-first platform services with explicit mobile and desktop implementations.

| Dependency/service                | Observed platform metadata            | Action                                                                             |
|-----------------------------------|---------------------------------------|------------------------------------------------------------------------------------|
| stockfish 1.8.1                   | android, ios                          | Replace desktop path with UCI Process runner and bundled binaries.                 |
| sqlite 2.4.2                      | android, ios, macos                   | Adopt sqlite3/drift/sqlite_common_ffi for Windows/Linux; plan migration.           |
| receive_sharing_intent 1.8.1      | android, ios                          | Use desktop file associations, drag/drop, open dialog, and command-line file args. |
| onesignal_flutter 5.4.0           | android, ios                          | Disable for desktop or replace with OS notifications if truly needed.              |
| appsflyer_sdk 6.17.9 rc           | android, ios                          | Disable in desktop builds; attribution should be web/installer campaign based.     |
| app_tracking_transparency 2.0.6+1 | ios                                   | Mobile-only; remove from desktop startup.                                          |
| purchases_flutter 9.12.3          | android, ios, macos, web              | No Windows in local metadata; design desktop entitlements.                         |
| google_sign_in 7.2.0              | android, ios, macos, web              | No Windows in local metadata; use browser/OAuth flow for Windows.                  |
| shorebird_code_push 2.0.5         | no plugin platforms in local metadata | Verify desktop support before relying on it; likely use installer/updater instead. |
| terminate_restart 1.1.0           | android, ios, web                     | Replace desktop update/restart behavior.                                           |

### Startup refactor

| Current concern                                                 | Desktop change                                                                                                             |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| main.dart initializes many services in one path.                | Split into AppBootstrap with platform service registry. Keep core services blocking; make optional services no-op or lazy. |
| SystemChrome orientation/UI calls assume mobile.                | Guard or remove for desktop windows.                                                                                       |
| RevenueCat API keys are hardcoded by iOS/Android branch.        | Move entitlements into a platform adapter and remove mobile branches from desktop.                                         |
| Deep link and PGN file intake use mobile sharing flows.         | Add desktop launch arguments, file associations, drag/drop, and recent documents.                                          |
| Hot restart handling is tuned for mobile engine/realtime leaks. | Keep useful cleanup, but separate desktop engine/process lifecycle.                                                        |

## 9. Roadmap

| Phase                | Duration  | Deliverables                                                                                                            | Exit criteria                                                                                         |
|----------------------|-----------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 0. Desktop proof     | 2-3 weeks | Compile/run a minimal desktop target; platform service registry; disabled mobile-only startup; one board with PGN open. | A developer can open a PGN on macOS/Windows target without mobile plugins crashing.                   |
| 1. Core extraction   | 3-4 weeks | core_chess package/module; PGN/notation tests; command model; workspace state model.                                    | Chess game tree, import/export, variation navigation are covered by tests and used by mobile/desktop. |
| 2. Desktop shell     | 4-6 weeks | Menu bar, command palette, shortcuts, docked panes, document tabs, board/notation/editor prototype.                     | Daily analysis session works keyboard-first for one game.                                             |
| 3. Local database    | 6-8 weeks | Desktop SQLite/FTS schema, import workers, search tables, duplicate detection, folder sync queue.                       | Import 100k+ PGN games locally and search/open without blocking UI.                                   |
| 4. Engine manager    | 4-6 weeks | UCI process runner, bundled Stockfish, profiles, MultiPV pane, cache, cancellation.                                     | Long-running analysis remains stable across document switches and app close.                          |
| 5. Professional prep | 6-8 weeks | Reference search pane, player dossier MVP, opening report MVP, blunder report MVP.                                      | A titled player can prepare for an opponent using Chessever without leaving the app.                  |
| 6. Beta hardening    | 4-6 weeks | Windows installer/signing, crash telemetry, desktop tests, performance budgets, update path, docs.                      | Private beta ready for coaches/pro players on Windows.                                                |

### First 30 days

| Week | Focus                            | Concrete tasks                                                                                                                                          |
|------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Make desktop boot boring         | Add platform service registry, no-op mobile-only services on desktop, guard SystemChrome/orientation, verify macOS target and define Windows CI runner. |
| 2    | Replace the two biggest blockers | Prototype sqlite3/drift or sqflite_common_ffl; prototype UCI Process runner with bundled Stockfish; add smoke tests.                                    |
| 3    | Workspace prototype              | Build desktop shell route with board, notation, engine pane, Gamebase pane, menu commands, and shortcut map.                                            |
| 4    | Data import MVP                  | Desktop file open/drag/drop, PGN import report, local games table, position index spike, performance measurements.                                      |

### Definition of a credible desktop beta

| Requirement | Threshold                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------|
| Performance | Open a game under 300 ms from local database, navigate moves under 16 ms/frame, import large PGN batches in background. |
| Keyboard    | All core analysis, navigation, search, save, engine, and annotation actions have shortcuts and menu entries.            |
| Database    | Local-first games, folders, search, duplicate detection, and Gamebase/reference search coexist clearly.                 |
| Engine      | Bundled Stockfish-compatible engine works without mobile plugin dependency and can be replaced by user engines.         |
| Reliability | No mobile-only plugin crashes on desktop; startup is resilient without network; recovery for corrupted local state.     |
| Packaging   | Signed Windows installer, app icon/resources, versioning, crash reporting, and update story.                            |

## 10. Risk Register

| Risk                                                              | Severity    | Mitigation                                                                                                                 |
|-------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------|
| Underestimating local database/index work                         | High        | Treat data_index as a first-class subsystem; benchmark with realistic PGN scale before polishing UI.                       |
| Mobile UI carried into desktop                                    | High        | Create a separate desktop shell and command system. Reuse widgets only after desktop interaction review.                   |
| Engine lifecycle instability                                      | High        | Use process-based UCI on desktop with cancellation, logs, and clean shutdown tests.                                        |
| Revenue/auth mismatch on Windows                                  | Medium-high | Decide entitlement provider early. Build OAuth/browser flow and license cache before beta.                                 |
| Legal/product scope around ChessBase proprietary formats/features | Medium-high | Support open formats first, especially PGN/FEN/UCI. Avoid reverse-engineering proprietary formats unless counsel approves. |
| Gamebase backend scale not enough for pro expectations            | Medium-high | Define data coverage target, update cadence, indexing plan, and cache warming strategy.                                    |
| Cross-platform ambition slows Windows launch                      | Medium      | Ship Windows-first professional beta; keep macOS/Linux architecture clean but not release-blocking.                        |
| Test coverage remains mobile-only                                 | Medium      | Add desktop widget/golden/integration tests and a Windows build job immediately after desktop boot.                        |

## Open product decisions

| Decision                                  | Why it matters                                                                                                          |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Windows-only first or simultaneous macOS? | ChessBase is Windows-centered; Windows-first reduces scope. macOS can follow if platform abstractions are clean.        |
| Local database format                     | Determines sync, import speed, FTS, migrations, and future data portability.                                            |
| Entitlement provider for desktop          | RevenueCat local metadata lacks Windows; payment/licensing affects startup and offline use.                             |
| Gamebase data coverage target             | Competing professionally requires a credible game count, update cadence, annotation strategy, and player identity data. |
| Desktop design language                   | Decide whether ChessEver becomes a dense pro tool, a modern coach/trainer, or both through modes.                       |

## Appendix A. Direct Dependency Readiness

Generated from current pubspec.lock and local pub-cache plugin metadata. A package without plugin platforms can still be usable on desktop if it is pure Dart/UI; the note calls out what should be validated.

| Package                    | Version       | Group                | Desktop readiness note                                                  |
|----------------------------|---------------|----------------------|-------------------------------------------------------------------------|
| amplitude_flutter          | 4.3.9         | observability/growth | Has desktop platform metadata: android, ios, macos, web                 |
| app_links                  | 6.4.1         | backend/network      | Has desktop platform metadata: android, ios, linux, macos, web, windows |
| app_settings               | 7.0.0         | platform/mobile      | Has desktop platform metadata: android, ios, macos                      |
| app_tracking_transparency  | 2.0.6+1       | observability/growth | Mobile-only. Hide behind platform adapter or remove from desktop build. |
| appsflyer_sdk              | 6.17.9+1-rc 1 | observability/growth | Mobile-only. Hide behind platform adapter or remove from desktop build. |
| build_runner               | 2.9.0         | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| cached_network_image       | 3.4.1         | other                | Dart/UI package; validate behavior on desktop.                          |
| chess                      | 0.8.1         | chess                | Dart/UI package; validate behavior on desktop.                          |
| chessground                | 9.0.0         | chess                | Dart/UI package; validate behavior on desktop.                          |
| clarity_flutter            | 1.4.2         | observability/growth | Dart/UI package; validate behavior on desktop.                          |
| collection                 | 1.19.1        | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| country_code               | 1.1.0         | i18n/assets          | Dart/UI package; validate behavior on desktop.                          |
| country_flags              | 4.1.2         | i18n/assets          | Dart/UI package; validate behavior on desktop.                          |
| country_picker             | 2.0.27        | i18n/assets          | Dart/UI package; validate behavior on desktop.                          |
| crypto                     | 3.0.6         | monetization/auth    | Dart/UI package; validate behavior on desktop.                          |
| cupertino_icons            | 1.0.8         | i18n/assets          | Dart/UI package; validate behavior on desktop.                          |
| dart_mappable              | 4.6.1         | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| dart_mappable_builder      | 4.6.1         | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| dartchess                  | 0.12.3        | chess                | Dart/UI package; validate behavior on desktop.                          |
| dio                        | 5.9.0         | backend/network      | Dart/UI package; validate behavior on desktop.                          |
| dropdown_button2           | 2.3.9         | state/ui             | Dart/UI package; validate behavior on desktop.                          |
| easy_debounce              | 2.0.3         | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| equatable                  | 2.0.7         | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| fast_immutable_collections | 11.0.4        | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| file_picker                | 8.3.7         | storage/files        | Has desktop platform metadata: android, ios, web, macos, windows, linux |
| firebase_core              | 3.15.2        | backend/network      | Has desktop platform metadata: android, ios, macos, web, windows        |
| flutter                    | 0.0.0         | other                | Dart/UI package; validate behavior on desktop.                          |
| flutter_animate            | 4.5.2         | state/ui             | Dart/UI package; validate behavior on desktop.                          |
| flutter_country_flags      | 0.0.1         | i18n/assets          | Dart/UI package; validate behavior on desktop.                          |
| flutter_dotenv             | 5.2.1         | i18n/assets          | Dart/UI package; validate behavior on desktop.                          |
| flutter_hooks              | 0.21.3+1      | state/ui             | Dart/UI package; validate behavior on desktop.                          |
| flutter_launcher_icons     | 0.14.4        | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| flutter_lints              | 5.0.0         | modeling/build/test  | Dart/UI package; validate behavior on desktop.                          |
| flutter_localizations      | 0.0.0         | i18n/assets          | Dart/UI package; validate behavior on desktop.                          |
| flutter_native_splash      | 2.4.7         | platform/mobile      | Plugin metadata is not Windows-ready: android, ios, web                 |

| Package                    | Version | Group                | Desktop readiness note                                                     |
|----------------------------|---------|----------------------|----------------------------------------------------------------------------|
| flutter_soloud             | 4.0.2   | i18n/assets          | Has desktop platform metadata: android, ios, linux, macos, windows, web    |
| flutter_svg                | 2.2.4   | i18n/assets          | Dart/UI package; validate behavior on desktop.                             |
| flutter_test               | 0.0.0   | modeling/build/test  | Dart/UI package; validate behavior on desktop.                             |
| font_awesome_flutter       | 10.10.0 | i18n/assets          | Dart/UI package; validate behavior on desktop.                             |
| gal                        | 2.3.2   | storage/files        | Has desktop platform metadata: android, ios, macos, windows                |
| google_sign_in             | 7.2.0   | monetization/auth    | No Windows plugin in local metadata. Desktop entitlement/auth path needed. |
| heroine                    | 0.7.2   | state/ui             | Dart/UI package; validate behavior on desktop.                             |
| hooks_riverpod             | 2.6.1   | state/ui             | Dart/UI package; validate behavior on desktop.                             |
| http                       | 1.5.0   | backend/network      | Dart/UI package; validate behavior on desktop.                             |
| icons_launcher             | 3.1.0   | modeling/build/test  | Dart/UI package; validate behavior on desktop.                             |
| image                      | 4.8.0   | storage/files        | Dart/UI package; validate behavior on desktop.                             |
| in_app_review              | 2.0.11  | platform/mobile      | Has desktop platform metadata: android, ios, macos                         |
| intl                       | 0.20.2  | i18n/assets          | Dart/UI package; validate behavior on desktop.                             |
| logarte                    | 1.4.0   | observability/growth | Dart/UI package; validate behavior on desktop.                             |
| logger                     | 2.6.2   | observability/growth | Dart/UI package; validate behavior on desktop.                             |
| marionette_flutter         | 0.3.0   | modeling/build/test  | Dart/UI package; validate behavior on desktop.                             |
| motor                      | 1.1.0   | state/ui             | Dart/UI package; validate behavior on desktop.                             |
| onesignal_flutter          | 5.4.0   | observability/growth | Mobile-only. Hide behind platform adapter or remove from desktop build.    |
| package_info_plus          | 8.3.1   | i18n/assets          | Has desktop platform metadata: android, ios, linux, macos, web, windows    |
| path                       | 1.9.1   | storage/files        | Dart/UI package; validate behavior on desktop.                             |
| path_provider              | 2.1.5   | storage/files        | Has desktop platform metadata: android, ios, linux, macos, windows         |
| patrol                     | 4.3.0   | modeling/build/test  | Has desktop platform metadata: android, ios, macos, web                    |
| patrol_finders             | 3.1.0   | modeling/build/test  | Dart/UI package; validate behavior on desktop.                             |
| permission_handler         | 12.0.1  | platform/mobile      | Has desktop platform metadata: android, ios, web, windows                  |
| purchases_flutter          | 9.12.3  | monetization/auth    | No Windows plugin in local metadata. Desktop entitlement/auth path needed. |
| receive_sharing_intent     | 1.8.1   | storage/files        | Mobile-only. Hide behind platform adapter or remove from desktop build.    |
| riverpod_annotation        | 2.6.1   | modeling/build/test  | Dart/UI package; validate behavior on desktop.                             |
| screenshot                 | 3.0.0   | storage/files        | Dart/UI package; validate behavior on desktop.                             |
| scrollable_positioned_list | 0.3.8   | state/ui             | Dart/UI package; validate behavior on desktop.                             |
| sentry_flutter             | 9.18.0  | observability/growth | Has desktop platform metadata: android, ios, macos, web, linux, windows    |
| share_plus                 | 10.1.4  | storage/files        | Has desktop platform metadata: android, ios, linux, macos, web, windows    |
| shared_preferences         | 2.5.4   | storage/files        | Has desktop platform metadata: android, ios, linux, macos, web, windows    |
| shorebird_code_push        | 2.0.5   | platform/mobile      | Dart/UI package; validate behavior on desktop.                             |
| showcaseview               | 5.0.1   | state/ui             | Dart/UI package; validate behavior on desktop.                             |
| sign_in_with_apple         | 7.0.1   | monetization/auth    | Has desktop platform metadata: android, ios, macos, web                    |
| skeletonizer               | 2.1.3   | state/ui             | Dart/UI package; validate behavior on desktop.                             |

| Package               | Version | Group           | Desktop readiness note                                                         |
|-----------------------|---------|-----------------|--------------------------------------------------------------------------------|
| smooth_page_indicator | 1.2.1   | state/ui        | Dart/UI package; validate behavior on desktop.                                 |
| smooth_sheets         | 0.15.0  | state/ui        | Dart/UI package; validate behavior on desktop.                                 |
| sprung                | 3.0.1   | other           | Dart/UI package; validate behavior on desktop.                                 |
| sqflite               | 2.4.2   | storage/files   | No Windows/Linux plugin. Move to sqflite3/drift or sqflite_common_ffi.         |
| stockfish             | 1.8.1   | chess           | Mobile plugin only in local metadata. Replace with desktop UCI process runner. |
| supabase_flutter      | 2.12.4  | backend/network | Dart/UI package; validate behavior on desktop.                                 |
| terminate_restart     | 1.1.0   | platform/mobile | Plugin metadata is not Windows-ready: android, ios, web                        |
| upgrader              | 12.1.0  | platform/mobile | Dart/UI package; validate behavior on desktop.                                 |
| url_launcher          | 6.3.2   | backend/network | Has desktop platform metadata: android, ios, linux, macos, web, windows        |

## Appendix B. Source Evidence

| Source                                              | Why it was inspected                                                                                |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| pubspec.yaml                                        | Project version, Dart SDK constraint, direct dependency declarations, assets, fonts, Patrol config. |
| pubspec.lock                                        | Resolved direct dependency versions.                                                                |
| .dart_tool/package_config.json + pub-cache pubspecs | Observed plugin platform metadata for desktop readiness.                                            |
| lib/main.dart                                       | Startup, service initialization, routes, app lifecycle, orientation, analytics/push/deep links.     |
| lib/repository/sqlite/app_database.dart             | Current local SQLite cache/key-value/list implementation.                                           |
| lib/repository/gamebase/gamebase_repository.dart    | Gamebase move aggregates, player/game search, eval, FEN normalization.                              |
| lib/screens/chessboard/analysis/chess_game.dart     | PGN parsing into game tree, clocks, eval comments, variations.                                      |
| lib/screens/chessboard/notation/notation_tree.dart  | Notation tree builder and PGN export.                                                               |
| lib/services/pgn_file_intake_service.dart           | PGN file/text intake and routing.                                                                   |
| lib/repository/library/library_repository.dart      | Supabase folders, saved analyses, shared books, subscriptions.                                      |
| lib/utils/responsive_helper.dart                    | Phone/tablet scaling assumptions.                                                                   |
| lib/widgets/tablet/tablet_layout_wrapper.dart       | Existing tablet split/content helpers.                                                              |
| test/ and patrol_test/                              | Current unit/widget and mobile E2E coverage.                                                        |

### Official ChessBase sources consulted

| Source                                                | Facts used                                                                                                                  |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| ChessBase '26 Mega Package                            | Windows requirements, Nov 2025 publication, opening report/reference search positioning, package contents, pricing context. |
| Mega Database 2026                                    | 11.7M+ games, 1475-2025 coverage, 114k+ annotated games, weekly update service, player lexicon scale.                       |
| ChessBase 18: New dimensions in study and preparation | Style report, blunder report, Lichess/Chess.com access, strategic themes, cloud engine, mobile/cloud preparation.           |
| ChessBase 18 reference search article                 | Reference-search positioning for team/opponent prep and opening variation knowledge.                                        |

[https://shop.chessbase.com/en/products/chessbase\\_26\\_mega\\_package](https://shop.chessbase.com/en/products/chessbase_26_mega_package)

[https://shop.chessbase.com/en/products/mega\\_database\\_2026](https://shop.chessbase.com/en/products/mega_database_2026)

<https://en.chessbase.com/newsroom/post/chessbase-18-new-dimensions-in-study-and-preparation?page=0>

<https://en.chessbase.com/post/chessbase-18-new-functions-in-the-reference-search>

End of document.